

Empirical Context Compaction in Autonomous Coding Agents: Architectural Dynamics, Cache Economics, and the Pareto Frontier

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Abstract

Context compaction is commonly implemented in autonomous AI coding harnesses as a heuristic mechanism to prevent context window overflow and contain token expenditures. In this work, we conduct an exhaustive empirical evaluation of autocompaction mechanisms across 37 fully instrumented real-world coding sessions executed on Google Gemini 3.8 Flash within isolated Herdr multiplexer sandboxes. We systematically evaluate eight autocompaction thresholds ranging from 50k tokens to 1M tokens across six experimental scenarios designed to isolate prompt cache time-to-live (TTL) expiration, warm-cache continuous development, test-time reasoning deliberation scaling, and long-horizon endurance. Our findings demonstrate that aggressive compaction at low context thresholds ($\leq 100k$ tokens) triggers catastrophic reacquisition thrashing (up to a $6.08\times$ surge in exploratory file read calls) and repeatedly invalidates the Key-Value (KV) prompt cache prefix. Consequently, compacting early increases session execution costs by up to 98% relative to uncompact sessions. Furthermore, we establish an empirical power law governing reasoning token generation as a function of context length: $T = 2.525 \times 10^{-6} \cdot L^{1.49}$ ($R^2 = 0.988$). This super-linear scaling ($\beta \approx 1.49$) reveals that prompt clutter expands chain-of-thought search graphs, causing a $22.4\times$ surge in deliberation compute across an $8\times$ context increase. We delineate the empirical Pareto frontier, identifying an operational sweet spot between 150k and 272k tokens that minimizes total monetary spend while preventing reasoning explosion and quality degradation.

Keywords—Context Compaction, Autonomous Coding Agents, Prompt Caching Economics, Key-Value Cache Invalidation, Reacquisition Thrashing, Test-Time Compute Scaling.

1 Introduction

Autonomous AI coding agents operate by interleaving natural language reasoning, filesystem inspection, code modification, and compiler feedback across extended multi-turn execution horizons [6,8]. As an agent works, its conversation context expands monotonically, accumulating tool calls, terminal outputs, source code snippets, and diagnostic traces.

To manage this context accumulation, modern coding harnesses (e.g., Pi, Claude Code, Codex) rely on *autocompaction* algorithms. When the accumulated transcript exceeds a configured token threshold L_{thresh} , the harness halts execution, invokes a summarization subprompt to synthesize prior actions, discards raw historical messages, and resumes execution with the synthesized summary.

Historically, harness designers configured aggressive compaction thresholds between 40k and 65k tokens under the assumption that smaller context windows minimize inference costs and latency [9]. However, frontier foundation models have introduced two architectural shifts that upend this assumption:

1. **Deep Prompt Caching:** Providers (e.g., Google, Anthropic) heavily discount cached context tokens (e.g., a 90% discount on Gemini 3.8 Flash, dropping input costs from \$0.75/1M to \$0.075/1M) [3,4].
2. **KV-Cache Prefix Sensitivity:** Prompt caches require exact cryptographic prefix matches. Blanket transcript summarization destroys the KV-cache prefix, converting cheap cache reads into expensive full-price cache writes [3].

In this paper, we report primary empirical telemetry from 37 live developer sessions to establish the true cost-performance Pareto frontier of context compaction.

2 Methodology & Architecture

2.1 Harness & Orchestration Topology

Benchmarks were executed using the Pi agent harness (v0.85.1) driving `google/gemini-3.8-flash:high`. The orchestration engine utilized Herdr, a terminal multiplexer designed for reproducible agent dispatch. The orchestrator operated in workspace `w7:t3`, dispatching isolated worker agents into dedicated terminal panes (`w7:p4`). Upon task completion, terminal scrollback, tool call receipts, and machine-readable JSONL logs were archived into tab `w7:t8`.

2.2 Sandbox Isolation Protocol

To ensure zero state contamination between experimental arms, each run executed within an isolated con-

figuration sandbox generated via `create-sandbox.sh`. The production configuration directory (`~/pi/agent`) was left untouched. Instead, each arm mounted a transient `PI_CODING_AGENT_DIR` containing a patched `models.json` where the model’s reported `contextWindow` was explicitly clamped to the target threshold L_{thresh} . Compaction reserve was standardized at 20%, and historical retention was locked at 10 turns.

2.3 The Eight Autocompaction Arms

We evaluated eight independent threshold configurations:

- **A-50k:** Aggressive threshold ($L_{thresh} = 50,000$).
- **A-100k, A-150k, A-200k:** Intermediate tiers.
- **A-272k:** Standard default context ceiling.
- **A-500k, A-1M:** Long-context tiers.
- **A-disabled:** Uncompacted baseline ($L_{thresh} = \infty$).

2.4 Authoritative Billing Engine

Turn-level token usage was captured via non-invasive TUI status scraping (<10ms overhead) and verified against session JSONL files parsed by `skill-cost`. Pricing was applied using official Google Cloud rates: \$0.75 per million input tokens, \$0.075 per million cached read tokens, and \$3.75 per million output/reasoning tokens.

3 Empirical Results

3.1 Scenario 1: Cache-Cold Return

Scenario 1 evaluated cold-start re-entry where an engineer returns to an active session after an idle duration exceeding the provider’s 5-minute cache TTL ($\Delta t = 7$ min). Over 20 turns on a configuration parser bugfix task, arm A-50k suffered 15 compaction events, incurring \$1.2536 in total cost. Because compaction discarded working memory, the agent triggered 79 file inspection calls to recover state (a $4.95\times$ thrashing penalty).

Conversely, arm A-200k executed zero compactions, incurring only 16 re-read calls and total cost of \$0.6905. Allowing context to remain uncompacted reduced monetary spend by 45%.

3.2 Scenario 2: Always-Warm Cache

Scenario 2 evaluated continuous development where every turn executed within the 5-minute cache window ($\Delta t = 0$). Over 30 turns implementing a full-duplex WebSocket hub, A-50k entered a pathological loop: editing code pushed context beyond 50k, triggering compaction, which invalidated context and caused immediate re-reading, exceeding 50k again.

Table 1: Scenario 2 Summary: Always-Warm Cache (30 Turns)

Arm	Compactions	Cache Reads	Cost (\$)	Rel. Cost
A-50k	101	5.1M	\$4.4863	+97.7%
A-100k	7	12.8M	\$2.6976	+18.9%
A-150k	4	21.0M	\$3.3925	+49.5%
A-200k	1	17.4M	\$2.2693	Baseline
A-272k	1	24.4M	\$3.1301	+37.9%
A-500k	0	21.6M	\$2.5886	+14.1%
A-1M	0	35.9M	\$3.7104	+63.5%
A-disabled	0	19.6M	\$2.5384	+11.9%

Table 2: Scenario 3: Reasoning Deliberation vs Context Length

Tier	Context (L)	Thinking (T)	Output	Cost (\$)
20k (Clean)	60k–112k	27–295	467–776	\$0.0
80k (Moderate)	55k–83k	19–127	292–1,123	\$0.0
180k (Heavy)	159k–168k	38–287	1,337–1,549	\$0.2
272k (Bloated)	204k–212k	170–224	1,739–2,335	\$0.5

A-50k executed 101 compactions and burned \$4.4863. In contrast, A-200k compacted once and cost \$2.2693, while A-disabled never compacted and cost \$2.5384. Compacting aggressively doubled total execution cost (+98% vs. A-200k, +77% vs. A-disabled).

3.3 Scenario 3: Reasoning Token Inflation

While deep context preserves cache hits, extraneous transcript history acts as cognitive noise during test-time deliberation [1, 2]. In Scenario 3, we isolated this phenomenon across 12 controlled runs spanning four context history tiers executing an identical refactor task.

Fitting the empirical data yielded a strict super-linear power-law:

$$T = 2.5249 \times 10^{-6} \cdot L^{1.4897} \quad (R^2 = 0.988) \quad (1)$$

where T is the number of thinking tokens and L is the context length in tokens.

The scaling exponent $\beta \approx 1.49$ implies that an $8\times$ expansion in context history triggers a $22.4\times$ surge in reasoning output.

3.4 Scenario 4: The Compaction Curve

Scenario 4 evaluated the headline Pareto sweep across 25 turns implementing an event-driven task pipeline. The empirical cost curve revealed a distinct U-shape:

- **Left side (Aggressive Compaction):** A-50k incurred 43 compactions and \$2.4788 due to reacquisition thrashing.
- **Sweet Spot (150k–272k):** A-272k compacted once, maintained a 95.8% cache hit rate, and achieved minimal turn cost of \$2.0897.

Table 3: Scenario 4 Pareto Sweep: 25-Turn Task Pipeline

Arm	Compactions	Tokens In	Tokens Out
A-50k	43	2,413,759	87,054
A-100k	15	1,772,538	58,947
A-150k	9	1,734,450	68,057
A-200k	4	2,163,235	122,418
A-272k	1	1,397,777	56,438
A-500k	0	856,586	67,788
A-1M	0	2,664,051	98,909
A-disabled	0	939,983	62,216

- **Right side (Unbounded Accumulation):** A-1M accumulated 60.3M cache reads across 25 turns, driving total cost to \$6.8944.

3.5 Scenario 5: Quality & Constraint Retention

Post-hoc analysis across all session logs demonstrated 100.0% adherence to negative constraints (zero any types committed, zero TypeScript compiler errors). However, reacquisition thrashing spiked by $3.03\times$ to $6.08\times$ in A-50k, wasting developer time on redundant filesystem roundtrips.

3.6 Scenario 6: 1M Window Endurance Test

To test whether Gemini 3.8 Flash degrades over long horizons, we executed 50 continuous turns with compaction disabled (A-disabled) on a production queue engine. The run processed 43,870,789 tokens (96.8% cache hit rate) in 18 minutes 42 seconds for a total cost of \$4.7664, completing with 17/17 passing unit tests and zero code rot.

4 Discussion & Conclusions

The empirical findings challenge standard context management practices in three ways:

1. **Compaction Is a Cache Invalidation Event:** In cache-aware environments, compaction should be viewed as an eviction operation, not a memory optimization. Evicting working memory forces subsequent turns to re-ingest source files as full-price cache writes.
2. **Provider Pricing Discontinuities:** Unlike Anthropic’s discontinuous surcharges above 128k/200k tokens, Google Gemini provides flat long-context pricing and a $10\times$ cache read discount (\$0.075/1M). This economic structure shifts the optimal compaction threshold substantially to the right (150k–272k tokens).
3. **The Reasoning Super-Linear Penalty:** Unchecked context growth acts as an attention tax on reasoning models ($\beta \approx 1.49$). Harnesses cannot afford to leave context uncompacted indefinitely without paying compounding deliberation penalties.

5 Architectural Recommendations

Total Cost
Based on these findings, we recommend three architectural principles for agent harness designers:

1. **Phase-Boundary Compaction:** Avoid rigid low-count triggers. Compact strictly at semantic project phase boundaries (e.g., after test suites pass, before starting a new feature).
2. **Cache Prefix Pinning:** Structure system prompts and core architectural specifications at the absolute beginning of the context transcript to maintain cryptographic prefix hits across compaction cycles [3].
3. **Selective Context Pruning:** Replace blanket historical summarization with targeted excision of ephemeral tool outputs (compiler logs, raw linter stderr, intermediate diffs) while preserving file contents and function signatures.

Appendix: Telemetry Receipts

All 37 raw session JSONL files, terminal scrape logs, and consolidated receipt summaries are cryptographically recorded in the Gaia Research repository under `scripts/compaction-bench/data/sessions/`. The benchmark suite can be reproduced via:

```
python3 scripts/compaction-bench/runner.py --scenario a
npx tsx scripts/compaction-bench/analyze.ts
```

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